

Midpoints and segment bisectors



1 No

2 It is the midpoint of $\overline{BD}$.

3 Point K

4

a Point G

b $\overline{EF}$

5 $PR = 5.1$ mm

6 $AB = 51.4$ cm

7

a $x = 9$

b $ML = 51$

8

a $k = 10$

b $XZ = 96$

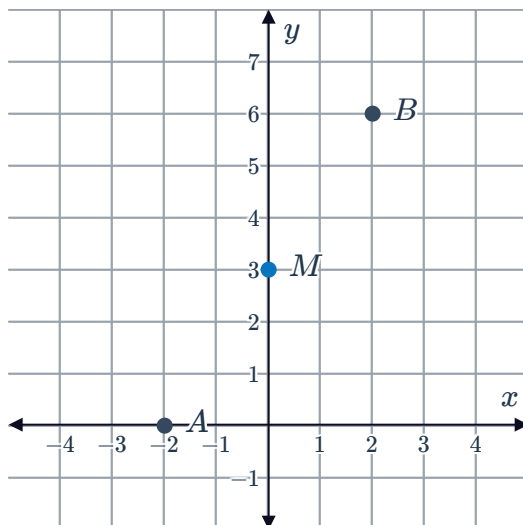
9 $\left(\frac{x_2 + x_1}{2}, \frac{y_2 + y_1}{2} \right)$

10

a 0

b 3

c



11

a $\left(\frac{1}{2}, \frac{5}{2}\right)$

c $\left(\frac{5}{3}, \frac{1}{10}\right)$



b $(-11, -8)$

d $\left(\frac{7}{2}, -\frac{7}{2}\right)$

12

a $(17, 10)$

c $(9, -6)$

b $(-3, -17)$

d $(11, -12)$

13 $(-6m, -4n)$

14 $(25^\circ \text{N}, 97^\circ \text{E})$

15 \$28 million

16 58 years

17

a 293 319

b 374 632

18 $PQ = 25$

19 $UV = 16x + 10$

20 $CD = 4\frac{2}{5}$ inches

21

a $\left(-\frac{15}{2}, 1\right)$

c $\left(-\frac{21}{2}, 11\right)$

b $(-9, 6)$

22

a $-2m - 4$

b $\frac{19n}{18}$

23

a $(2, 5)$

b $(-1, -11)$



24

a $p = -\frac{3}{2}$

b $q = 0$

Additional questions

25 $M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$

26

a $M(5, -2)$

b $M(10, 6)$

27

a $M(5, 7)$

b $M\left(-\frac{11}{2}, -8\right)$

28

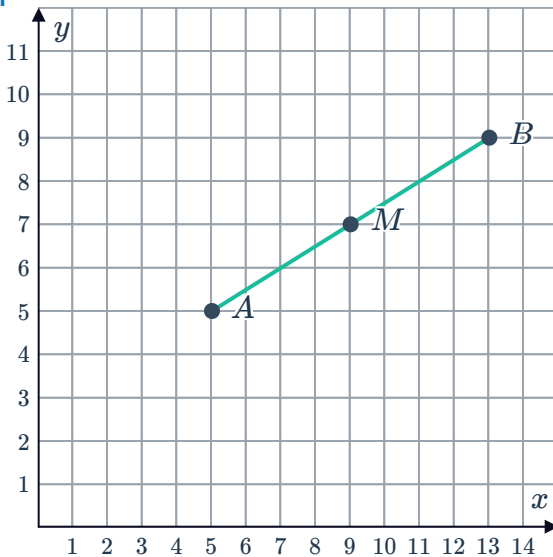
a $M(-3, 2)$

b $M(-3, -2)$

29

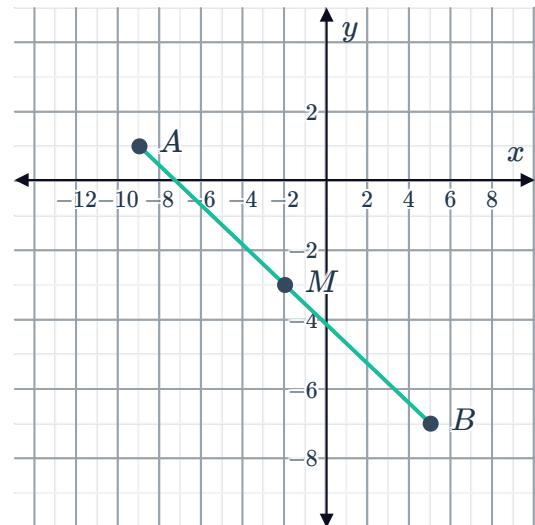
a i $M(9, 7)$

ii



b i $M(-2, -3)$

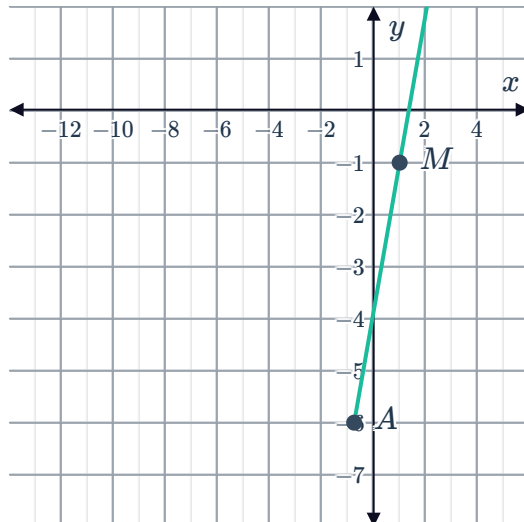
ii



c

i $M(1, -1)$

ii

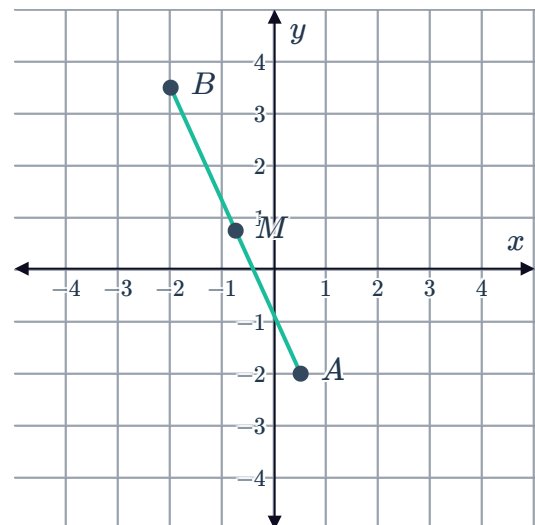


B



i $M\left(-\frac{3}{4}, \frac{3}{4}\right)$

ii



30 $(4m, 3n)$

31

a $A(5, 6)$

c $A(-12, -8)$

b $A(4, -3)$

d $A(-4, -5)$

32 $(28^\circ \text{N}, 83^\circ \text{E})$

33 $Q(-6, 1), R(-8, 4), S(-10, 7)$

34

a $p = -\frac{1}{4}$

b $q = 0$

35

a $\overline{KL} \cong \overline{JK}, \overline{HL} \cong \overline{HI}, \text{ and } \overline{EI} \cong \overline{IJ}.$

b $\overline{CD}$ bisects $\overline{JL}$ and $\overline{IL}.$

There are multiple equivalent names for $\overline{CD}$ and $\overline{GM}.$

c Point K is the midpoint of $\overline{LJ}$ and Point I is the midpoint of $\overline{EJ}.$

36 D is the midpoint of $\overline{CF}, E$ is the midpoint of $\overline{DF}$ and $\overline{CB},$ and F is the midpoint of $\overline{DB}.$